UTOPI-AR: User Behavior Guided Token Pre-Compress for Efficient Egocentric Long-Video Understanding in AR Environment

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Abstract

As long-video understanding gains research traction, Augmented Reality (AR) devices stand out as a practical deployment platform. However, heavy redundancy in egocentric video creates substantial compute and storage challenges on AR hardware. Therefore, we introduce UTOPI-AR, a plug-and-play pre-compress module for egocentric long videos tailored to AR platforms: First, we leverage user motion signals to estimate inter-frame overlap and introduce a viewpoint-shift-invariant pruning algorithm. Second, we exploit eye-tracking sensor to extract user fixations and separate attended foreground from background for preferential retention. Moreover, we present a new egocentric video benchmark that leverages user behavior to better simulate real human questioning patterns, and further explore the performance of different models and efficient algorithms on it. Across multiple benchmarks and model backbones, our UTOPI-AR achieves over 95% *compression rate while maintaining the same performance.*

1. Introduction

Augmented Reality (AR) is emerging not only as a novel user interface but also as a machine learning (ML) platform that integrates rich embodied experience such as sensing, perception, memory, and speech. By aligning the digital and physical realms, AR enables real-time extraction of useful information and contextual decision-making, transforming domains such as healthcare [15, 23, 36], education [3, 46, 49], and industry [16]. Beyond immersive applications, AR devices provide continuous multimodal sensory data streams for ML research. Equipped with cameras, eye-tracking sensors, and motion sensors, they capture high-bandwidth visual (e.g. egocentric video stream), pose, and behavioral streams in real time. These heterogeneous, high-throughput signals create challenges and opportunities for long-context ML models to capture correlations between these signals, attention, and the environment over extended timescales.

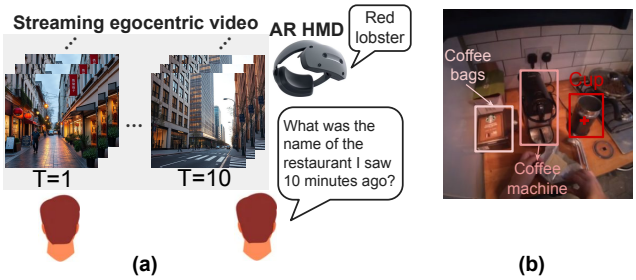


Figure 1. (a) An example on real-life AR LEU scenario. (b) An example illustrating how user attention varies with gaze location.

As shown in Figure 1 (a), consider an AR head-mounted display (HMD) used during driving navigation. The device continuously captures and encodes multimodal inputs with ML models, and a brief user glance at a restaurant must be reliably bound to the corresponding visual features. Later, answering a query such as "What was the name of the restaurant we passed 10 minutes ago?" requires an episodic memory module that can index temporal embeddings, retrieve the correct event, and interface with a downstream processing engine (e.g., a large model). This capability of *long video understanding* (LVU) transforms AR from a passive sensing platform into an intelligent personal assistant with long-context reasoning, marking a critical step toward next-generation AR applications. In education, students could revisit past demonstrations or experiments to strengthen understanding; in industrial settings, recorded task sequences could be replayed to identify errors; in healthcare, surgeons could query specific moments when key anatomical landmarks appeared; and in every-

day life, AR could serve as a personal memory aid, helping users relocate items such as keys.

However, this LVU functionality poses stringent demands on resource-limited AR hardware. To illustrate the scale of the challenge, consider an AR headset that records high-definition video at 30 frames per second: even with compression, this can generate several gigabytes of data per hour, and at higher resolutions and frame rates, such as 4K at 60 frames per second, the storage demand quickly rises to tens of gigabytes per hour, far exceeding the practical capacity of wearable devices. Second, continuously running encoder backbones for online encoding, retrieval, and reasoning is computationally and energetically prohibitive on AR hardware, necessitating new algorithms that performs LVU with high better computational and storage efficiency.

To address the first challenge, we propose a perception guided approach to improve LVU efficiency in AR environments, inspired by human attentional mechanisms. Human perception is inherently nonuniform, with fidelity highest at the locus of attention and gradually decreasing with eccentricity [5, 22, 43]. In egocentric AR video, most content is closely tied to the user's current task, and previous studies [9, 25, 55] indicate that human intent is strongly correlated with the questions posed and their corresponding answers, resulting in substantial spatial redundancy in unattended regions. Additionally, Egocentric AR streams also exhibit strong temporal redundancy, since head and body motion are typically smooth and structured rather than random, so large portions of consecutive frames contain overlapping areas, as shown in Figure 5. Our key idea is to exploit these spatial and temporal regularities by prioritizing storage and computation around behaviorally salient regions and events, thereby reducing memory footprint and processing cost while preserving information essential for long horizon understanding.

Building on these observations, we introduce *UTOPI-AR*, a plug and play **U**ser behavior guided **TO**ken **P**re compress module **I**n **AR** environments. UTOPI-AR improves efficiency by exploiting spatial redundancy in unattended regions, adaptively discarding unimportant tokens based on gaze informed attention from the user. When gaze signals are unavailable, we employ a machine learning based attention predictor to estimate the most likely attended locations, and UTOPI-AR prunes redundant visual information accordingly to reduce both storage and computational cost. In addition, for temporal redundancy, we retain the complete information of the overlap region only at its first appearance in the sequence. In subsequent frames, the corresponding areas are replaced with their compressed token representations.

Finally, while multiple benchmarks have been proposed for LVU, they primarily focus on generic video based queries and do not fully capture the **human centric** and **attention guided** nature of real AR usage, where questions are closely tied to user attention patterns [9, 25, 55]. Motivated by this gap, we develop a new benchmark, termed *EgoEverything*, for LVU that simulates realistic, attention driven interactions in AR environments. Overall, our contribution can be summarized as follows:

- We introduce UTOPI-AR, a plug-and-play user behavior guided token pre-compress module that uses AR native signals to compress visual tokens before query stage. It is capable of compressing both temporal and spatial redundancies, significantly reducing inference and storage costs on AR devices.
- We propose EgoEverything, a egocentric video understanding benchmark using multi-agent generation pipeline and user behavior inspired sampling, producing both attention driven and detail oriented QA pairs.
- Across multiple egocentric LVU benchmark and diverse baseline model, our approach achieve consistent efficiency–accuracy benefits: maintaining baseline accuracy while achieving substantial token compression (up to 95% on EgoSchema) and markedly reducing on-device storage.

2. Background and Related Work

2.1. Vision Language Models

Contemporary vision–language models (VLMs), including proprietary systems [39, 47] and open source counterparts [6, 18, 57], extend language based foundation models to visual and multimodal inputs and now exhibit strong, broadly general capabilities. Trained at scale on spatially rich data [13], they deliver high accuracy in spatial reasoning while implicitly encoding human preferences and priors. By unifying vision and language within a single framework, these models support perception, reasoning, instruction following, and dialogue, enabling AR assistants that ground semantics directly in what the user sees and interacts with.

2.2. Long-context Egocentric Video Understanding

Long-context Egocentric Video Understanding has become a key focus in modern machine learning, fueled by large scale first person recordings that continuously capture how people move, look, and interact in everyday settings. For AR systems, these streams offer a persistent record of the user's visual and behavioral context, enabling assistance that can refer back to what has happened minutes, hours, or days earlier in complex real world environments. A central trend in this area is the use of structured long term memories rather than treating video as a flat sequence [4, 8, 10, 17, 48, 51, 52]. In egocentric vision, prior work has organized continuous recordings into task or activity centric threads [21, 42, 51] and constructed egocentric

scene graphs that encode objects, agents, and their evolving relations from the wearer's viewpoint [24, 28, 44]. These structured memories can then be selectively retrieved in response to natural language queries and supplied to downstream multimodal models, such as VLMs, which operate over the retrieved subset to produce grounded, temporally informed answers.

2.3. Long video understanding Efficiency

Most existing methods either sample temporal frame like VidF4 [31] and GenS [54] or select query-relevant spatial regions based on the user's question like DynTok [58] and Q-Frame [59], yet these approaches all require user queries as input. Our UTOPI-AR can perform pre-compression before the query stage, which significantly reduces data storage costs and do not need to reprocess the video for each query. Some approaches like DTD [53] and PruneVid [27], perform temporal and spatial compression without relying on queries, yet these methods do not take into account users' behavior and gaze information. In contrast, UTOPI-AR leverages user behavior and gaze information more effectively, enabling more efficient visual-token pruning in AR/VR scenarios.

2.4. Attention-Guided Question Formation in Egocentric AR

Human perception operates selectively rather than uniformly across the visual field. Classic studies describe visual attention as a "spotlight" [20, 41], which in practice is often approximated by gaze position. The allocation of attention typically decreases with distance from this focus and can be modeled with functions such as a two dimensional Gaussian [29], yielding highest fidelity near the attended location and progressively lower fidelity in the periphery [12, 19].

This selectivity has direct implications for the kinds of questions an AR user will ask. As illustrated in Figure 1(b), a user wearing an AR HMD in the kitchen may fixate on a cup on the counter; this object, having received strong attentional focus, is a natural candidate for later queries such as "Where did I leave the cup?". Nearby objects with weaker attentional weighting, depicted with lighter bounding boxes in Figure 1(b), are correspondingly less likely to be referenced. Cognitive psychology and vision science studies similarly report that gaze patterns are predictive of subsequent memory and information seeking behavior [30, 56].

2.5. AR Hardware System

Figure 2 illustrates a representative AR hardware configuration. Typical headsets integrate multiple front and side facing cameras that capture both the surrounding environment and aspects of user state, including gaze direction. Outward facing cameras produce high resolution views of

Figure 2. (a) Meta Aria glasses equipped with multiple cameras. (b) Meta Quest Pro headset, which functions as both an AR and VR device.

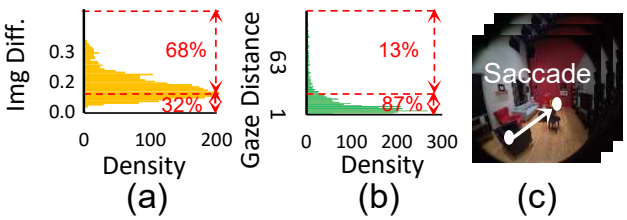


Figure 3. (a) Distribution of the image differences. (b) Distribution of the gaze distances. (c) Video Fragments in the user's view.

the scene (for example, 1408×1408 on the Meta Aria Glass [2]), while inward facing cameras record lower resolution monochrome images of the eyes. Together, these sensing channels support gaze estimation [1, 2, 38] using either analytic models such as pupil–corneal reflection or modern learning based approaches [33], providing essential signals of user attention and interaction for many AR applications. However, certain AR glasses do not include inward facing cameras or native gaze tracking, in which case additional machine learning based mechanisms are required to infer user attention from outward facing video and other sensor signals.

In addition, dedicated tracking cameras support simultaneous localization and mapping (SLAM), which, when tightly integrated with inertial data from the onboard IMU, enable robust visual–inertial odometry and full six-degree-of-freedom head-pose estimation, in line with pipelines such as ORB-SLAM3 [11]. During this process, SLAM continuously reconstructs a sparse 3D map of the environment composed of map points, which are stable visual features that are repeatedly observed across frames and serve as persistent anchors for camera localization. Combined, this sensing stack delivers high-quality scene imagery, attention cues, and accurate pose tracking, making modern AR devices well suited for fine-grained, intention-aware segmentation and interaction.

2.6. Human Attention in AR Environment

Human eye movement consists of three main behaviors: *fixations*, *saccades*, and *smooth pursuit*. During a fixation, the eyes remain relatively still, focusing on a single point to capture detailed visual information, and the fixation location often reflects where a person's attention is directed.

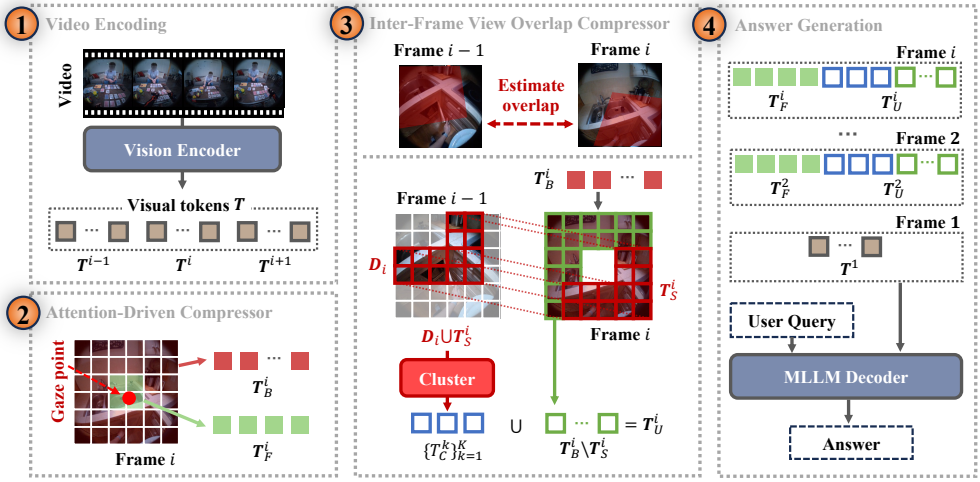


Figure 4. The overall architecture of UTOPI-AR.

Saccades are extremely fast movements that rapidly shift the gaze from one fixation point to another, allowing the visual system to explore the scene efficiently within milliseconds. In contrast, Smooth Pursuit movements, which enable the eyes to follow a moving object continuously, occur less frequently and mainly when tracking visible motion. Together, these coordinated actions reveal both the perceptual process and the attentional focus that support efficient understanding of dynamic environments.

To investigate the attention patterns of AR users, we adopt the *ARIA Everyday* (AEA) dataset [34], which was collected using the Meta Aria smart glasses. To collect this dataset, more than 150 participants wore the AR device during their everyday activities across hundreds of recording sessions, resulting in over 1,000 hours of egocentric multimodal data. The dataset captures synchronized first-person video, gaze trajectories and head motion, providing a comprehensive view of natural human behavior. By analyzing these recordings, we can quantify how users allocate attention in real-world environments and examine how fixation density, saccadic frequency, and pursuit events reveal underlying visual and cognitive patterns. Specifically, we aim to investigate two aspects: (1) how temporal redundancy manifests across consecutive video frames over time, and (2) how user attention, represented by gaze locations, is spatially distributed within each AR frame at every timestep.

As shown in Figure 3, we measure the RGB pixel-wise differences between consecutive frames to quantify visual changes over time. The results show that approximately 32% of consecutive frames exhibit less than 9% change in pixel values, indicating a high degree of temporal redundancy and strong correlation between adjacent frames. This observation suggests that much of the visual content in egocentric AR videos remains stable across short time intervals, providing opportunities for efficient processing and computation reuse.

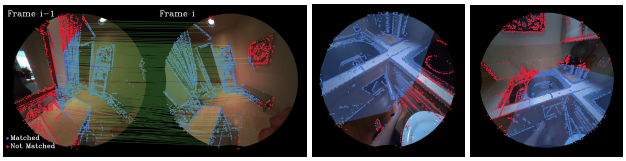
Moreover, we group consecutive video frames into clusters when their RGB differences are below a predefined threshold ϵ. For each frame cluster, we calculate the spatial distance between fixation gaze locations to assess attention stability. As shown in Figure 3, the histogram of fixation gaze distances indicates that 87% of frames within each cluster exhibit minimal gaze shifts ($\leq$ 22 pixels), suggesting that AR users consistently focus on one or two salient objects within each cluster. These focused regions are likely related to the users' subsequent questions and answers in the LVU task, further highlighting a strong spatial redundancy within short temporal segments of AR video sequences.

3. Methodology

3.1. UTOPI-AR Overview

Most multimodal large language models (MLLMs) process a video by first sampling frames to obtain frame set $\mathbf{I}$, then encoding $\mathbf{I}$ with a ViT-based vision encoder to get visual tokens $\mathbf{T} = \{\mathbf{T}^i\}_{i=1}^N, \mathbf{T}^i = \{T_{w,h}^i | w \in [1, W], h \in [1, H]\}$. Here, i denotes the index (i.e., timestamp) of the frame $\mathbf{I}^i$ from which $\mathbf{T}^i$ originates. N is the number of sampled frames. (w, h) is the spacial location of $T_{w,h}^i$ in $\mathbf{I}^i$, $H \times W$ is the size of the token map. Next, the positional embeddings $P(\cdot)$ are applied to $\mathbf{T}$ to obtain $\mathbf{T}_{emb} = P(\mathbf{T})$. Note that we adopt QWen2.5VL [7] as the MLLM baseline, which uses a RoPE-style [45] positional encoding. Finally, $\mathbf{T}_{emb}$ is fed together with textual tokens into the MLLM decoder to generate the final answer. To reduce computational and storage cost on AR devices, our goal is to compress the number of tokens in $\mathbf{T}_{emb}$ and store them on-device for subsequent user queries.

The overall UTOPI-AR pipeline, shown in Figure 4, contains four parts: *Video Encoding, Attention-Driven Com-*



(a) (b) frame $i-1$ (c) frame i

Figure 5. Illustration of Inter-frame overlap and SLAM map point.

pressor (ADC), Inter-Frame View Overlap Compressor (IVOC) and Answer Generation. After getting visual tokens $\mathbf{T}$ for all frames from video encoding step, we first apply the ADC in each frame $\mathbf{I}^i$. It utilize real-time gaze point to separates foreground tokens $\mathbf{T}_F^i \subset \mathbf{T}^i$ that are highly related to the user's activity and close to the gaze point. The rest are background tokens $\mathbf{T}_B^i = \mathbf{T}^i \setminus \mathbf{T}_F^i$. We then use the IVOC to further remove temporal redundancy within $\mathbf{T}_B^i$, producing $\mathbf{T}_U^i$. The final compressed visual token set is $\mathbf{T}' = \mathbf{T}^1 \cup \{P(\mathbf{T}_F^i) \cup P(\mathbf{T}_U^i)\}_{i=2}^N$. Next, we will elaborate implementation details of AD-compressor and IVO-compressor.

3.2. Attention-Driven Compressor

As noted in Section 2.5, AR devices include an eye-tracking module that records the 2D coordinates of the user's gaze point (g_x^i, g_y^i) for each frame $\mathbf{I}^i$. Given the set of visual tokens $\mathbf{T}^i$, we compute the distance between each token $T_{w,h}^i \in \mathbf{T}^i$ and the gaze point of its corresponding frame, and select tokens within a distance threshold ρ as the foreground tokens:

$$\mathbf{T}_F^i = \{T_{h,w}^i | (w - g_x^i)^2 + (h - g_y^i)^2 < \rho\}. \tag{1}$$

The positional indices (i, w, h) of each foreground token $\mathbf{T}_f^i$ are kept identical to the original, preserving spatiotemporal structure of foreground tokens.

For AR glasses that lack built-in gaze-tracking capabilities, we employ the deep learning–based SUM method [26] to estimate the user's gaze location on each egocentric frame, serving as a proxy for human attention. Specifically, for each frame $\mathbf{I}^i$, the SUM outputs a saliency map $\mathbf{S}^i$. Trained extensively on egocentric video datasets with ground-truth gaze points, SUM lets us interpret the saliency score $\mathbf{S}^i(x, y)$ at location (x, y) as the probability that the user's attention is focused there. In our implementation, we take the gaze point for frame $\mathbf{I}_i$ as

$$(g_x^i, g_y^i) = \arg\max_{(x,y)} \mathbf{S}^i(x, y). \tag{2}$$

Subsequent procedures remain the same as described above.

3.3. Inter-Frame View Overlap Compressor

Because egocentric videos are typically captured as a single continuous stream, they exhibit strong scene continuity, as illustrated in Figure 3(c). Prior work (e.g., DTD [53]) exploits this property by comparing the similarity of visual tokens at each spatial location over time and discarding redundant tokens, thereby significantly reducing both the input size to the MLLM and its associated storage and computation costs.

However, in egocentric AR video streams, tokens representing the same content often shift spatially due to changes in the user's head pose. For instance, Figure 5b and Figure 5c illustrate two consecutive frames from the AEA dataset at times $i-1$ and i, where the blue region highlights the area visible in both frames. We observe that corresponding visual tokens generally reappear at different spatial locations rather than remaining aligned at the same position.

As described in Section 2.5, an AR device records user head pose (position and orientation) information during operation. With the camera parameters, we can project the SLAM map points into the screen space to obtain their pixel locations. We then identify the map points that are visible in both frames to estimate the inter-frame overlap region between $\mathbf{I}^i$ and $\mathbf{I}^{i+1}$, as illustrated in Figure 5a. Each dot is a SLAM map point: red dots appear only in the current frame, and blue dots appear in both frames. After finding matched points between two frames, we can estimate the coordinate mapping of the overlap region between the frames, denoted by $f(\cdot, \cdot)$, which can map a coordinates of $\mathbf{I}^i$ to $\mathbf{I}^{i-1}$, as shown in Figure 5a by green lines.

For the overlapped visual area, we keep only one full copy in an earlier frame and replace it in later frames with compressed **surrogate tokens**. Concretely, we first obtain the background tokens $\mathbf{T}_B^i$ of frame $\mathbf{I}^i$, Then, identify the inter-frame overlap region $\mathbf{O}^i$ between frame $\mathbf{I}^i$ and $\mathbf{I}^{i-1}$ and its corresponding tokens $\mathbf{T}_O^i = \{T_{w,h}^i | (w, h) \in \mathbf{O}^i\}$. We then use $f(\cdot, \cdot)$ to map coordinates of $\mathbf{T}_O^i$ into $\mathbf{I}^{i-1}$ and compute the cosine similarity with the tokens at the corresponding locations:

$$s_{w,h}^i = \mathrm{cosim}\big(T_{f(w,h)}^{i-1},\ T_{w,h}^i\big), \tag{3}$$

where $\mathrm{cosim}(\cdot, \cdot)$ denotes cosine similarity. Next step we detects tokens inside $\mathbf{O}^i$ that do not changed sufficiently compared to $\mathbf{I}^{i-1}$:

$$\mathbf{T}_S^i = \big\{T_{w,h}^i \in \mathbf{T}_O^i \mid s_{w,h}^i > \theta\big\}, \tag{4}$$

where θ is the similarity threshold. Next, We find counterparts of $\mathbf{T}_S^i$ in the previous frame $\mathbf{I}^{i-1}$,

$$\mathbf{D}_i = \big\{T_{f(w,h)}^{i-1} \mid s_{w,h}^i > \theta,\ (w, h) \in \mathbf{O}^i\big\}, \tag{5}$$

and cluster these tokens $\mathbf{D}_i \cup \mathbf{T}_S^i$. This yields K cluster centers $\{T_C^k\}_{k=1}^K$, which we use as surrogate tokens to replace $\mathbf{T}_S^i$.

Notably, for each surrogate token T_C^k, we set its temporal position to the current frame index i, and its spatial position

to the mean of the spatial position of all tokens in its cluster, and mapped back to $\mathbf{I}^i$ via the inverse coordinate mapping $f^{-1}(\cdot,\cdot)$. Finally, we obtain:

$$\mathbf{T}_U^i = \mathbf{T}_B^i \setminus \mathbf{T}_S^i \cup \{T_C^k\}_{k=1}^K. \tag{6}$$

Additionally, for datasets that do not include SLAM map points or user-motion data, we apply image matching using LightGlue [32], a lightweight deep neural network designed to learn robust local feature correspondences across images.

3.4. EgoEverything Collection

Finally, while multiple benchmarks have been proposed for LVU, they primarily focus on generic video based queries and do not fully capture the **human centric** and **attention guided** nature of real AR usage, where questions are closely tied to user attention patterns [9, 25, 55]. Motivated by this gap and the background presented in Section 2.6, we develop a new benchmark, termed *EgoEverything*, for LVU that simulates realistic, attention driven interactions in AR environments.

EgoEverything is constructed from real-world egocentric video datasets featuring authentic gaze tracking, including the Aria Everyday Activities (AEA) [34] and Nymeria [35] datasets. The AEA dataset contains 143 clips totaling 7.3 hours of annotated daily activities captured across five indoor environments with synchronized gaze data, while Nymeria comprises approximately 300 hours of recordings collected from nearly 50 indoor and outdoor scenes with real gaze traces. Each video is divided into semantically coherent segments by clustering frame embeddings. For each segment, a one-sentence action summary is generated by a vision–language model and verified by human annotators; the concatenation of these summaries forms the contextual basis for question construction.

Building on the video summarization results, we introduce an *attention-guided object sampling module* that selects questioning subjects by simulating the human attention mechanism described in Section 2.4. This module first detects all objects within each video segment, leveraging a VLM to extract their bounding boxes and semantic labels. For each frame, it then computes the distance between the gaze point and the center of each detected object's bounding box. Based on this information, the module randomly selects a *Target Object* from the segment and, using the perceptual sampling function $P_\theta(\cdot)$, chooses a single object within that frame as the Target Object. A simple implementation of $P_\theta(\cdot)$ can be modeled as a 2D Gaussian distribution, defined as:

$$P_\theta(.) \propto e^{-\frac{||d-g||^2}{2\theta^2}} \tag{7}$$

where g is the gaze location and $P_\theta(\cdot)$ specifies the sampling frequency at object center d, which decays as the distance $\|d-g\|$ increases. The parameter θ controls the decay behavior of the sampling function. Multiple Target Objects are sampled and subsequently used for the question generation process.

To generate questions confronting human questioning patterns, two VLM agents are employed for question generation and review. The Generation Agent (GA) is responsible for formulating multiple-choice questions (MCQs) about the selected target object, while the Reviewing Agent (RA) inspects these questions and provides feedback. After the VQA dataset is generated, the questions are further refined through over **300 hours** of manual inspection and rule-based filtering.

4. Experiment

4.1. Evaluation Setups and Implementation Details

Video-QA Benchmarks To evaluate the effectiveness of UTOPI-AR, we use three public video question–answering benchmarks: EgoSchema [37], EgoGazeVQA [40], and our EgoEverything. Because UTOPI-AR targets AR scenarios, all selected datasets are built on Egocentric videos and multiple-choice questions. EgoSchema provides over 5k automatically constructed five-choice questions and offers a broad test of long-context understanding, reasoning, and retrieval. EegoGazeVQA supplies user-gaze traces, with questions categorized into causal, spatial, and temporal reasoning. Our EgoEverything contains over 5k multiple-choice QA pairs (examples in the Appendix). Unlike template-based generation, our questions are produced by AI agents conditioned on realistic AR usage contexts and human-attention signals, yielding queries that follow natural human questioning patterns and provide a more faithful, human-like evaluation setting for long-context egocentric video understanding in AR environments.

Implementation Details. We implement our method on the Qwen2.5VL-7B [6] models and use 5 NVIDIA A6000(48GB) GPUS for all experiments. For video input, we uniformly sample 360 frames per video for all experiments conducted on EgoEverything and EgoSchema, for experiments on EgoGazeVQA, we sample a total of 100 frames per video, giving priority to keyframes that provide gaze point. To ensure a consistent token budget across all experiments for fair performance comparison, we set different values of θ and ρ for each experimental configuration, as detailed in the Appendix. In our experiments, the EgoEverything dataset provides real-time user gaze data, as well as IMU and SLAM map point information. For EgoSchema, since both of these modalities are unavailable, we employ the SUM [26] model and LightGlue [32] to predict user gaze points and inter-frame overlap in advance. For EgoGazeVQA, we directly use its provided native gaze points and utilize LightGlue to estimate inter-frame overlap. In the implementation of the PruneVid [27], we dis-

Table 1. Accuracy Evaluation of UTOPI-AR

Method	Token Budget	EE	ES	EGV
QWen2.5VL-7B	100%	55.0%	60.1%	47.8%
QWen2.5VL-7B	0%*	36.6%	49.4%	43.4%
QWen2.5VL-7B				
w/ UTOPI-AR (ours)	15%	**57.3%**	**64.8%**	**49.0%**
w/ DTD [53]	15%	53.0%	57.5%	47.2%
w/ PruneVid [27]	15%	51.3%	63.0%	48.8%
QWen2.5VL-7B	15%†	50.4%	53.8%	44.2%
MindPalace [28]	15%†	33.2%	45.2%	37.9%
PLLaVA [50]	15%†	34.1%	29.6%	31.5%
Videollama3-7b [57]	15%†	46.4%	49.7%	42.5%
InternVL2.5-8B [14]	15%†	52.4%	51.5%	43.0%
QWen2.5VL-7B				
w/ UTOPI-AR (ours)	50%	**57.1%**	**61.9%**	**48.5%**
w/ DTD	50%	51.5%	60.2%	45.2%
w/ PruneVid	50%	52.4%	60.4%	47.8%
w/ UTOPI-AR (ours)	30%	**57.4%**	**62.6%**	**48.7%**
w/ DTD	30%	53.6%	60.1%	42.8%
w/ PruneVid	30%	53.3%	61.9%	48.1%

able its KV-cache compression module because it requires modification of QWen2.5VL and some details are not open sourced. For the MindPalace [28], we adopt QWen2.5VL as the backbone LLM for generating final answers.

4.2. Performance on Benchmarks

In Table 1, we evaluate various VLM baselines and combinations of QWen2.5VL with different compression methods on three benchmarks: EgoEverything (**EE**), EgoSchema (**ES**), and EgoGazeVQA (**EGV**). All methods are tested under the same token budgets. During the experiments, we use identical prompts for all models and feed the video, question, and answer options to the model in a fixed order. The *Token Budget* in the table refers to the ratio of the number of visual tokens after compression to that before compression.

We mark token budget with a "*" when the it is 0%, meaning that only the question and answer options are provided to the model, without any video input. This setting equivalent to the *blind test*. Some token budgets are marked with "†" to indicate that these models do not use any token compression method; instead, to control for token budget, we downsample the video resolution to ensure that the total number of visual tokens remains consistent.

Experimental results show that under the same three token budgets $(15\%, 30\%, 50\%)$, our proposed UTOPI-AR method consistently outperforms all other efficiency baselines across all benchmarks. Specifically, when compressing QWen2.5VL with UTOPI-AR, even at only 15% visual tokens retained, the accuracy improves by an average of 2.7% compared to using all visual tokens. Compared with other compression algorithms, UTOPI-AR achieves at least 2.1% higher accuracy on average. Moreover, relative to the naive uniform-resolution downsampling of QWen2.5VL, the average accuracy improvement of UTOPI-AR is 4.3%. Finally, compared with the blind test, UTOPI-AR uses only 15% of the visual tokens yet achieves an average accuracy gain of 14.2%, demonstrating the effectiveness of our approach.

4.3. Efficiency Analysis

To evaluate the performance of our plug-and-play method on different baselines and to analyze the trade-off between accuracy and token budget, we apply UTOPI-AR to the PLLaVA [50] model and compare its results under various token budgets on two benchmarks, as shown in Table 2. Here, a token budget of 100% denotes the vanilla baseline. The best accuracy is highlighted in bold, while the smallest token budget that maintains accuracy no lower than the vanilla baseline is marked with an underline.

From the experimental results, we observe that except for PLLaVA on EgoEverything, where applying UTOPI-AR causes a slight accuracy drop, compressing visual tokens with UTOPI-AR generally improves accuracy across other baselines and benchmarks. One possible reason is that our method removes interference from redundant tokens, allowing it to achieve better performance under smaller token budgets. Moreover, when using QWen2.5VL as the baseline, our method can compress up to 95% of the visual tokens without any loss in accuracy. Applied to PLLaVA on the EgoSchema dataset, UTOPI-AR still achieves up to 85% token reduction while preserving performance. Even on EgoEverything, where a slight accuracy drop occurs, the decrease remains within 2%. These results show that our plug-and-play approach strikes an excellent balance between efficiency and accuracy, exhibiting strong robustness across different baselines and datasets. In Table 3, we evaluate the computational overhead of QWen2.5VL with and without UTOPI-AR. As shown, regardless of whether the AR device provides user gaze points or SLAM map point information, our method significantly accelerates the model, demonstrating its strong efficiency advantage.

4.4. Ablation Study

Module Ablation Study To validate the effectiveness of the ADC and IVOC modules, we conduct ablation studies, as shown in Table 5. We evaluate QWen2.5VL-7B with UTOPI-AR under different module configurations on the EgoSchema. The results show that enabling either module improves accuracy, with both achieving around 64%. When both modules are activated, the accuracy reaches its maximum of 64.8%, along with the highest compression ratio. Since the ADC and IVOC modules respectively address spatial and temporal redundancies, they can jointly enhance compression efficiency without negatively affect-

Table 2. Efficiency Analysis.

Baseline	Token Budget	EE	ES
PLLaVA	100%	**34.8%**	37.8%
w/ UTOPI-AR	50%	33.7%	39.8%
w/ UTOPI-AR	30%	34.2%	**40.1%**
w/ UTOPI-AR	15%	33.5%	39.3%
w/ UTOPI-AR	5%	31.8%	32.1%
QWen2.5VL-7B	100%	55.0%	60.1%
w/ UTOPI-AR	50%	57.1%	61.9%
w/ UTOPI-AR	30%	**57.4%**	62.6%
w/ UTOPI-AR	15%	57.3%	**64.8%**
w/ UTOPI-AR	5%	55.3%	61.4%
w/ UTOPI-AR	3%	50.5%	59.4%

Table 3. Overhead Analysis.

Token Budget	Overhead w/ AR signal	Overhead w/o AR signal
100% token	466 TFLOPS	466 TFLOPS
15% token	52 TFLOPS	87.4 TFLOPS

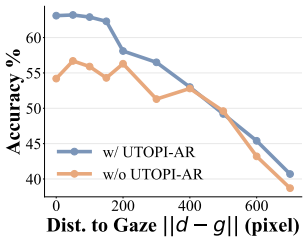
Table 4. Performance comparison on the Low-Motion video subset "SLOW" and the Motion-Intensive video subset "FAST".

Method	Token Budget	ES		
		FAST	SLOW	AVG
QWen2.5VL-7B w/UTOPI-AR	30%	62.0%	63.2%	62.6%
QWen2.5VL-7B w/UTOPI-AR	15%	63.6%	66.0%	64.8%
QWen2.5VL-7B w/UTOPI-AR	5%	59.2%	63.6%	61.4%

Table 5. Evaluation of different ADC/IVOC configurations

ADC	IVOC	Token Budget	ES
✓	✓	15%	64.8%
✓	×	50%	64.1%
×	✓	30%	64.4%
×	×	100%	60.1%

Figure 6. Impact of different "Question-Gaze" distance.



ing accuracy. This demonstrates that the two modules are complementary and can be effectively combined to achieve both higher compression and better overall performance.

Impact of Motion-Intensive Video To assess the robustness of the IVOC module, we compare QWen2.5VL-7B with UTOPI-AR on *Motion-Intensive* and *Low-Motion* videos (see Table 4). To construct the evaluation split, we first encode each EgoSchema video with the QWen2.5VL vision encoder and obtain a token map for every frame, denoted $\mathbf{T}^i$. We then average all token embeddings within each frame to get a single frame-level representation, $\mathbf{r}_i = \text{mean}(\mathbf{T}^i)$. Next, we compute the cosine similarity between consecutive frame representations and average over the video to obtain the mean inter-frame similarity

$$\alpha = \frac{1}{N}\sum_{i=1}^{N} \text{cosim}(\mathbf{r}_i, \mathbf{r}_{i+1}), \tag{8}$$

We sort all EgoSchema videos by α and select the top 50% (larger α) as *Low-Motion* videos (marked SLOW); the remaining 50% are *Motion-Intensive* videos (marked FAST). As shown in Table 4, under the same token budget, SLOW videos generally achieve higher accuracy than FAST. However, the gap is small, averaging 2.6%. This indicates that our IVOC module is robust across videos with different levels of temporal redundancy.

Impact of Object-Gaze Distance To verify the robustness of the ADC module, we group the questions in the Ego-Everything benchmark according to the distance between the question subject and the gaze point: $\|d-g\|$ (as defined in Equation 7). We then test the accuracy of QWen2.5VL-7B with and without UTOPI-AR under different distance groups, as shown in Figure 6. As $\|d-g\|$ increases, both models show a decrease in performance. This is because when the queried object is far from the gaze point, it usually appears near the edge of the video frame, where its appearance is less stable. Such questions are typically more challenging. It can be observed that when the question subject is close to the gaze point, our UTOPI-AR method shows a clear advantage, achieving about 10% higher accuracy. This improvement comes from the ADC module's ability to leverage user attention and preserve the most informative regions for video understanding. Even when the question subject is far from the gaze point, UTOPI-AR maintains similar accuracy to the vanilla version, indicating strong robustness. This is because even if the queried object falls outside the attended region, the IVOC module can still preserve its complete information when it first appears.

5. Conclusion

We propose UTOPI-AR, a training-free user behavior guided token pre compress module in AR environments. UTOPI-AR improves efficiency by using gaze-informed attention to discard spatially redundant visual tokens. When gaze is unavailable, it predicts likely attended regions and prunes accordingly to reduce storage and computation. Experiments show that UTOPI-AR can compress up to 95% of visual tokens on QWen2.5VL and up to 85% on PLLaVA while maintaining baseline accuracy. UTOPI-AR achieves the highest accuracy (64.8%) under the strongest compression and maintains robustness across motion levels, with only a 2.6% gap between low-motion and fast-motion

videos. Our method provides a practical and effective solution for improving long-video understanding efficiency in AR/VR scenarios.

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